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MIL-L-49148B(CR)
8 November 1989
SUPERSEDING
MIL-L-49148A(ER)
1 October 1980

MILITARY SPECIFICATION

LENS ASSEMBLY, OBJECTIVE VIEWER, DRIVER'S NIGHT VISION, AN/VVS-2(V)

This specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Objective Lens Assembly, herein known as objective lens, which is part of Viewer, Driver's Night Vision, AN/VVS-2(V).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-11268	-	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	-	Marking of Electronic Items
MIL-O-13830	-	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, USA Communications-Electronics Command ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703-5000 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-150	-	Photographic Lenses
MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-810	-	Environmental Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

SM-B-771334	-	Barrel Assembly, Lens, Objective
SM-C-771354	-	Retainer, Lens, Element 7 and 8

(Copies of Specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The objective lens assembly is mounted to and is part of the Viewer, Driver's Night Vision AN/VVS-2(V).

3.2 Construction. The objective lens assembly shall be constructed in accordance with SM-B-771334 and as specified herein.

3.3 Qualification. Furnished under this specification shall be product which are authorized by the Government as qualified products (see 3.10.3).

3.3.1 Initial production testing. The supplier shall furnish and test the objective lens as specified in the contract or purchase order (see 6.2).

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3.4 Parts and materials. Parts and materials shall be as specified herein and shown on applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

3.5 Performance characteristics.

3.5.1 Effective focal length. The objective lens shall have an Effective Focal Length (EFL) of 33.6, ±0.5 millimeters (mm).

3.5.2 Field of view. The field of view shall be not less than 44 degrees for a 25 mm diameter format.

3.5.3 Linear distortion. The objective lens shall have been 9 percent and 12 percent barrel distortion.

3.5.4 Modulation Transfer Function (MTF) on axis. The MTF shall be not less than the values specified in Table I.

TABLE I. Objective lens modulation transfer function.

Test Frequency (cycles/millimeter) on the Image Intensifier	Modulation Transfer (Percent)
10	84
20	70
30	55
40	40

3.5.5 T-number. The T-number shall be no greater than 1.25.

3.5.6 Veiling glare. The veiling glare shall be no greater than 2.5 percent.

3.5.7 Sealing. The objective lens shall be assembled and sealed in an atmosphere of dry nitrogen containing 1 mole-percent helium. The helium leak rate of the sealed objective lens shall be not greater than 9.9×10^{-4} cm^3/sec.

3.5.8 Flange focal length. The flange focal length shall be not less than 0.039 inches (0.991 mm) when measured from the 1.5-inch diameter central area of retainer (SM-C-771354).

3.6 Environmental.

3.6.1 High temperature. The objective lens shall withstand storage temperature of +68°C without damage or degradation of performance (see 3.10.1 or 3.10.5).

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3.6.2 Low temperature. The objective lens shall withstand storage temperature of -51°C without damage or degradation of performance (see 3.10.1 or 3.10.5).

3.6.3 Temperature shock. The objective lens shall withstand cycles of sudden changes in temperature between -51°C and +68°C without damage or degradation of performance (see 3.10.1 or 3.10.5).

3.6.4 Vibration. The objective lens shall withstand vibration equivalent to that encountered in tracked vehicles during 5000 miles of operation without damage or degradation of performance (see 3.10.1 or 3.10.5).

3.6.5 Shock. The objective lens shall withstand basic shock of 40 g's for 18 milliseconds (ms). The objective lens shall withstand simulated weapon firing shock of 100g's for 1.5 ms on half sine pulse without degradation of performance (see 3.10.5).

3.7 Treatment and painting. Unless otherwise specified, treatment and painting shall conform to MIL-STD-171.

3.8 Marking. Unless otherwise specified, the objective lens shall be marked and identified per MIL-M-13231.

3.9 Workmanship. Standards of workmanship shall be such that components will meet all requirements of this specification and any referenced specifications or drawings where inspected in accordance with Section 4. Parts and assemblies shall exhibit none of the defects categorized in MIL-O-13830.

3.9.1 Cleanliness. The objective lens shall meet the cleanliness requirement for optical systems as specified in MIL-O-13830.

3.10 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

3.10.1 Damage. Damage is defined as any breakage, loosening, shift, dents, failure of any finish, hardware, connection, or component, or condensation, fogging or moisture on any internal surface.

3.10.2 Ambient temperature. Ambient temperature is defined as +23°C, ± 10°C.

3.10.3 Qualified products. A product which has successfully met all of the requirements of an IPT or first article witnessed by the procuring activity.

3.10.4 Environmental tests. For the purpose of this specification, environmental tests are those tests discussed in 4.6.

3.10.5 Degradation of performance. Any deterioration in quality, or failure to operate with the requirements of this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. Initial production tests (4.3)
b. Quality conformance inspection (4.4)
c. Inspection of packaging (4.7)

4.3 Initial production testing.

4.3.1 Inspection. Each IPT objective lens shall be inspected in accordance with Table II, either in process or during assembly, as required. Presence of one or more defects shall be cause for rejection of that objective lens and may be cause for rejection of IPT.

TABLE II. Inspection.

Requirements	Requirement paragraph
Parts and materials not as specified	3.4
Treatment and painting	3.7
Identification or special marking missing or illegible	3.8
Workmanship	3.9
Cleanliness	3.9.1

4.3.2 Tests. Following successful completion of the inspections specified in 4.3.1, each IPT objective lenses shall be subjected to all tests listed in Table III. Tests may be performed in any order.

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Failure of any inspection shall be cause for rejection of that assembly and may be cause for rejection of IPT.

TABLE III. Initial production testing.

Inspection	Requirement paragraph	Test paragraph
Effective focal length	3.5.1	4.5.1
Field of view	3.5.2	4.5.2
Linear distortion	3.5.3	4.5.3
Modulation transfer function	3.5.4	4.5.4
T-number	3.5.5	4.5.5
Veiling glare	3.5.6	4.5.6
Sealing	3.5.7	4.5.7
Flange focal length	3.5.8	4.5.8
High temperature	3.6.1	4.6.1
Low temperature	3.6.2	4.6.2
Temperature shock	3.6.3	4.6.3
Vibration	3.6.4	4.6.4
Shock	3.6.5	4.6.5

4.3.3 Disposition of IPT samples. Disposition of IPT samples shall be in accordance with the contract or purchase order.

4.4 Quality conformance inspection.

4.4.1 Inspection. Each objective lens shall be inspected as specified in Table II. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests.

4.4.2.1 Group A inspection. Objective lenses that have passed the examination specified in 4.4.1 shall be tested in accordance with Table IV in the order listed. Failure of any test shall be cause for rejection of that unit. Assemblies which pass Group A inspection shall be formed into lots by the contractor.

TABLE IV. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Effective focal length	3.5.1	4.5.1
Field of view	3.5.2	4.5.2
Modulation transfer function	3.5.4	4.5.4
Sealing	3.5.7	4.5.7
Flange focal length	3.5.8	4.5.8

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4.4.2.2 Group B. Group B tests shall be conducted on objective lenses selected from units which have passed the Group A test. Sampling shall be in accordance with the contract or purchase order. Test shall be conducted in accordance with Table V.

TABLE V. Group B inspection.

Test	Requirement paragraph	Test paragraph
Linear distortion	3.5.3	4.5.3
T-number	3.5.5	4.5.5
Veiling glare	3.5.6	4.5.6

4.4.2.3 Group C. This inspection shall consist of the tests specified in Table VI and shall be performed on units that have been subjected to and have met Group A inspection. Sampling shall be as specified in the contract or purchase order.

4.4.2.3.1 Disposition of Group C samples. Objective lenses which have been subjected to Group C tests and are free from damage may be shipped as part of the contract.

TABLE VI. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
High temperature	3.6.1	4.6.1
Low temperature	3.6.2	4.6.2
Temperature shock	3.6.3	4.6.3
Vibration	3.6.4	4.6.4
Shock	3.6.5	4.6.5

4.4.2.3.2 Group C failures. Actions required in the event of a Group C failure shall be as specified in the contract or purchase order (see 6.2).

4.5 Test procedures. Unless otherwise specified all tests shall be performed at ambient temperature (see 3.10.2). Temperature chambers shall maintain specified temperatures within $\pm$ 2.0°C.

4.5.1 Effective focal length. The effective focal length of the objective lens shall be measured in accordance with the procedures of 5.1.2.2 of MIL-STD-150. Failure to meet the requirements of 3.5.1 shall constitute failure of this test.

4.5.2 Field of view. The field of view of the objective lens shall be measured in accordance with MIL-STD-150. Failure to meet the requirements of 3.5.2 shall constitute failure of this test.

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4.5.3 Linear distortion. The linear distortion of the objective lens shall be measured in accordance with MIL-STD-150, and the percent distortion shall be calculated with the following formula:

$$\text{Percent distortion} = \frac{F_A - F_I}{F_I} \times 100$$

Where:

F_A = Actual image height measured (mm)

F_I = Ideal image height (mm)

= EFL x tan θ

and:

EFL = Effective focal length (see 4.5.1)

$$= \frac{F_I\ (\theta\ 1)}{\tan \theta\ 1}$$

where $\theta\ 1 = 1^o$ and $F_I\ (\theta\ 1)$ is image height measured at $\theta\ 1 = 1^o$

Failure to meet the requirements of 3.5.3 shall constitute failure of this test.

4.5.4 Modulation transfer function. The objective lens shall be tested on axis with a sine wave pattern setup that utilizes an S-20ER response photomultiplier conforming to Figure 1 and a tungsten illumination source operated at 2854K, ±200 K color temperature. (If a microscope objective is used to relay the image of the objective lens, the numerical aperture (NA) of the microscope objective must be equal to or greater than 0.65). The pattern-collimator combination must allow the objective lens to be tested either continuously or at 8 distinct evenly distributed points from 0 to 40 line pairs per millimeter (lp/mm) including 0 lp/mm. The test shall be repeated with the objective rotated 90 degrees about the optical axis. Failure to meet requirements of 3.5.4 shall constitute failure of this test.

4.5.5 T-number. The T-number of the objective lens shall be measured in accordance with Method 5 or 6 of MIL-STD-150. The light source used shall be operated at 2854 K, ±200 K color temperature. The detector used shall have an S-20ER response (see Figure 1). Failure to meet the requirements of 3.5.5 shall constitute failure of this test.

4.5.6 Veiling glare. The stray light or veiling glare of the objective lens shall be measured in accordance with Method 35 of MIL-STD-150. The detector used in this procedure shall have an S-20ER response (see Figure 1). Failure to meet the requirements of 3.5.6 shall constitute failure of this test.

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4.5.7 Sealing. The objective lens shall be tested for helium leakage with a Varian Model 925-40 "Porta-Test" or equivalent. A leak rate greater than that specified in 3.5.7 shall constitute failure of this test.

4.5.8 Flange focal length. The flange focal length shall be measured as the distance from the flange as described in 3.5.8 to the aerial image through the lens formed by a collimated target. The aerial image shall be in the plane of best definition. This measurement shall be performed utilizing a microscope which is aligned to the reference flange. Failure to meet the requirements of 3.5.8 shall constitute failure of this test.

4.6 Environmental. The objective lens shall be retested to meet the requirements of 3.5.4, 3.5.6, and 3.5.7 after all operation and environmental testing has been completed. Failure to meet these requirements shall constitute failure as defined in degradation of performance environmental tests per 3.10.4.

4.6.1 High temperature. The objective lens shall be subjected to the test of Method 501.1, Procedure 1, of MIL-STD-810, except that temperature of Steps 2 and 3 (highest storage temperature) shall be +68°C. Failure to meet the requirements of 3.6.1 shall constitute failure of this test.

4.6.2 Low temperature. The objective lens shall be subjected to the test of Method 502.1, Procedure 1, of MIL-STD-810. The storage temperature (Step 2) shall be -51°C and shall be maintained until temperature stabilization of the test item is reached. Failure to meet the requirements of 3.6.2 shall constitute failure of this test.

4.6.3 Temperature shock. The objective lens shall be subjected to the test of Method 503.1, Procedure I, of MIL-STD-810. Immediately after each cold cycle, the assembly shall be examined for internal condensation. Evidence of moisture or condensation on any internal surface or failure to meet the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Vibration. The objective lens shall be subjected to MIL-STD-810, Method 514.2, Procedure VIII, Table 514.2-6, Tracked Vehicles, limited to one sweep in the vertical axis only. Failure to meet requirements of 3.6.4 shall constitute failure of this test.

4.6.5 Shock. The objective lens shall be subjected to the test of MIL-STD-810, Method 516.2. Failure to meet requirements of 3.6.5 shall constitute failure of this test.

a. 40g's at 18 milliseconds, Figure 516.2-1, shall be the basic design test, Shock Procedure I.

b. Weapon firings, 100g's using procedure IV except at 1.5 ms, Figure 516.2-2.

4.7 Inspection for packaging. Packaging shall be inspected in accordance with the requirements of Section 5.

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5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The objective lens is used with the AN/VVS-2(V) for driver viewing during periods of darkness.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Necessary actions by contractor in the event of a Group C failure (see 4.4.2.3.2).
c. Level A, Level B, or commercial preservation and packing (see Section 5).
d. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.
e. Sampling plans for Group B and Group C testing. As guidance, unless otherwise specified, sampling shall be conducted per the requirements of MIL-STD-105.
f. Quantity and schedule for first article testing for all line items and disposition of IPT samples.
g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).
h. Qualification - If product is not qualified at time of award the contract must require qualification prior to first delivery.
i. MIL-STD-810C shall be used for environmental tests of section 4.

6.3 Definitions. See 3.10.

6.4 Subject term keyword listing.
Lens Assembly
Objective Lens
Viewer Driver's

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the change.

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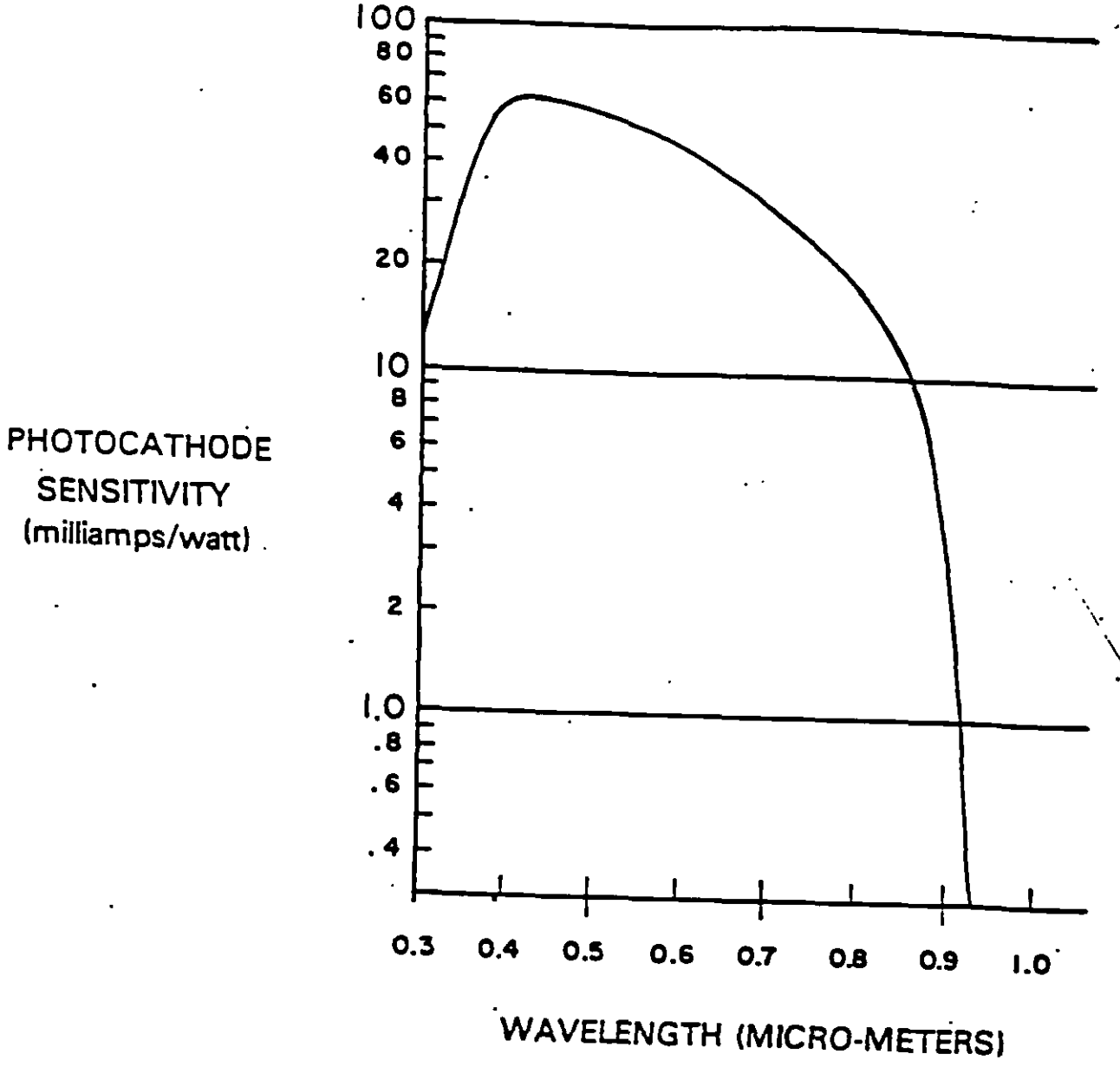


FIGURE 1. RESPONSE CURVE, S-20 EXTENDED RED PHOTOCATHODE

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-49148B(CR)	LENS ASSY, OBJ. VIEWER, DRIVER'S NIGHT VISION , AN/VVS-2(V)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
3b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
7b. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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